Full Stack Assignment – 38

Assignment ID: FSD-38

Title: Quota-Limited Request Management System

Duration: 3 Hours

1️⃣ Objective

Design a **Quota-Limited Request Management System** that allows users to make limited requests within a defined quota.

The system must enforce quota limits and persist all request data.

2️⃣ User Roles

USER

- Register and log in
- Submit requests
- View remaining quota

ADMIN

- Register and log in
- Define quota limits per user
- View request usage reports

3️⃣ Functional Requirements

Authentication & Authorization

- JWT authentication
- Role-based backend & frontend access

Quota Management

- ADMIN assigns quota limits
- USER requests consume quota
- Quota data stored in database

4️⃣ Business Rule (Critical)

⚠️ **Requests must be rejected once the assigned quota is exhausted.**

Quota validation must happen before processing requests.

5️⃣ Technical Constraints

- Database persistence mandatory
- No in-memory counters
- JWT + role validation required
- Secure environment config

6 Frontend Requirements

- Quota display for USER
- Request submission interface
- Quota management view for ADMIN

7 Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8 Evaluation Note

Evaluation focuses on **revocation enforcement, role control, and end-to-end execution**.